

Answer Key

For: All Quiz Questions Formatted PDF

Milestone 1

Q1	False
Q2	8
Q3	agn
Q4	(not A) or (not B)
Q5	[10, 20, 30, 40] [10, 20, 30, 40]
Q6	list
Q7	$O(n^2)$
Q8	As n grows, B will eventually take longer to run than A
Q9	$O(n)$ and $O(n^2)$
Q10	$O(n)$
Q11	increases by a factor of two
Q12	stays the same
Q13	best-case = average-case = worst-case
Q14	It doubles in size every time a new element is added to it
Q15	$O(1)$
Q16	By allocating a new larger array and copying existing elements over
Q17	the index of the first occurrence of element e in the array
Q18	i to n-1
Q19	$O(1)$
Q20	True

Stacks, Queues, Deques

Q1	Deque can be used either as a Stack or a Queue
Q2	The operations provided to manipulate the type
Q3	Queue -> FIFO, Stack -> LIFO
Q4	[10, 40]

Single Linked List

Q1	Data in arrays is stored in contiguous memory locations but not linked lists
Q2	$O(n)$
Q3	$O(1)$
Q4	Walk the entire list from the head, one step at a time ($O(n)$)

Double Linked List

Q1	is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list
Q2	is $O(1)$ in both a singly-linked list and a doubly-linked list
Q3	is $O(n)$ in both a singly-linked list and a doubly-linked list
Q4	is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list
Q5	is $O(1)$ in both a singly-linked list and a doubly-linked list

Preparation for Exam 2

Q1	8
Q2	[10, 20, 30, 40] [10, 20, 30, 40]
Q3	$O(n^2)$
Q4	increases by a factor of two
Q5	The operations provided to manipulate the type
Q6	pop_back
Q7	Queue -> FIFO, Stack -> LIFO
Q8	Deque can be used either as a Stack or a Queue
Q9	[10, 20, 20, 40]
Q10	[30, 40]
Q11	[40, 30, 40]
Q12	<code>node = Node(item, self.head); self.head = node; if self.tail is None: self.tail = node</code>
Q13	<code>curr.prev.next = curr.next ; curr.next.prev = curr.prev</code>
Q14	prints all but the last item in the list

Recursion

Q1	all recursive call chains end in a base case
Q2	It must have exactly one call to itself.
Q3	It eventually terminates with an error (exception)
Q4	It is almost always preferable to use recursion, if possible.
Q5	4

Recursion II

Q1	Returns the number of times 'a' occurs in the first n letters of a_str
Q2	It will recur endlessly and never return
Q3	In the recursive call we need to call bar(n-1) instead of bar(n)
Q4	It will recur endlessly and never return
Q5	We need to add a condition: if m == 0: return 1

Tree

Q1	Preorder
Q2	Postorder
Q3	3
Q4	2^n

Binary Search Tree

Q1	2 3 5 7 8 9
Q2	Left child on '5'
Q3	$O(\log n)$
Q4	$O(n)$
Q5	$O(n)$

Maps and Hash Functions

Q1	Dictionary
Q2	Hash table
Q3	Converting a key into an integer value in a fixed range

Maps and Hash Functions

Q4	Generation and compression of a hash code
Q5	n / N

Priority Queues

Q1	$O(\log n)$
Q2	$O(\log n)$
Q3	The rightmost element of the deepest level
Q4	Every level is fully filled, except possibly the deepest, which is filled left to right
Q5	A complete binary tree